

A General-Purpose Semi-Humanoid Robotic Platform

Technical Project Overview

Introduction, Objectives, and Design Perspective

1 Introduction

Autonomous robots deployed in service, laboratory, and light-industrial settings are almost always built for one job at a time: a fixed gripper for a fixed task, on a body that cannot be repurposed once that task changes. This project takes the opposite starting point. Rather than designing a robot around a single application, we are designing and building a **semi-humanoid robotic platform**: a wheeled mobile base, an anthropomorphic upper body with dual manipulators, and a perception head, integrated as one coordinated mechatronic system that is not committed to any single task.

Our contribution is not a new task-specific robot, but the mechanical, electrical, and software *integration layer* that makes such a body usable, reliable, and ready to be commanded – so that the work of the eventual user begins at the level of intent, not at the level of motor drivers.

2 Objectives

General Objective

To design, integrate, and deliver a modular semi-humanoid robotic platform – complete in mechanical structure, electrical systems, and software stack – that exposes its perception, navigation, and manipulation capabilities through a standardized, ready-to-use interface, and to validate this platform as a foundation for future research.

Specific Objectives

- Design a modular mechanical structure (mobile base, torso, arms, head) that supports independent development, maintenance, and future upgrade of each subsystem.
- Build a distributed electrical and embedded-control architecture that safely and reliably drives all actuators and sensors.
- Develop an integrated software stack – perception, navigation, and manipulation – built primarily on established open-source frameworks rather than developed from first principles.
- Provide a single, standardized command interface through which a user issues high-level orders and receives feedback, without needing to touch low-level control.
- Provide a dedicated software port through which a task-specific program can be loaded onto the platform, to demonstrate and validate the platform concept end to end.
- Treat **modularity** and **integration** as the two central engineering pillars of the project, above building any single subsystem from scratch.
- Document the design and integration process as a research paper, in keeping with the research-and-development nature of the project.

3 What Is a General-Purpose Platform?

A *platform*, in the sense used throughout this project, is the complete integration of a robot's mechanical structure, electrical systems, and software into one operational whole that exposes a fixed set of stable, high-level capabilities – locomotion, perception, and manipulation – rather than one fixed behavior. It is the finished, working body itself: everything below the level of decision-making, built, tested, and ready to run.

Calling the platform *general-purpose* means that no particular task is encoded into its mechanical structure or its low-level control loops. What the robot physically has – a mobile base that can navigate, a perception system that can see and localize objects, arms that can reach and grasp – stays exactly the same no matter what the robot is eventually used for. Only the decision-making layer above it changes: what to move to, what object to pick, when to act. Because that decision layer is kept separate from the platform, the same physical robot can support many different downstream applications, one after another, without any redesign of its hardware, actuation, or control stack – as long as each application expresses itself as a sequence of actions the platform already knows how to perform.

This is the core distinction we are building around: the platform provides the *capability*, and each application provides the *intent*.

4 User Interaction and the Task Port

A user does not command the platform by writing motion trajectories or tuning controllers – they issue it high-level orders through a standardized set of command interfaces (topics): move forward a given distance, navigate to a specified location, pick up a specified object, and so on. The platform is responsible for turning each such order into safe, coordinated motion – planning the path, perceiving the environment, and executing the manipulation – entirely on its own.

To prove that this separation genuinely works, the platform will expose a dedicated **software port**: an interface through which a user can flash a task-specific program onto the system. This program is intentionally kept outside the platform’s core stack – it interacts with the robot exclusively through the same standardized topics available to any user, requesting navigation, perception, or manipulation actions and reading back state. Successfully flashing and running one concrete task through this port, without modifying the platform itself, is our proof of concept: evidence that the platform’s interfaces are sufficient, on their own, to build a real application on top.

5 System Overview

Development spans three tightly coupled domains, unified by one design rule: modularity first, integration always.

- **Mechanical** – a wheeled base, torso and telescopic column, dual manipulators, and a perception head, each built as an independently serviceable, interchangeable module.
- **Electrical** – a tiered embedded architecture: a high-level compute unit for perception and decision-making, and dedicated microcontrollers closing real-time control loops per subsystem over a shared, standardized bus.
- **Software** – perception, navigation, and manipulation pipelines assembled from mature open-source robotics frameworks, integrated rather than reinvented, and exposed through one coherent set of high-level topics. We will also integrate a Vision-Language-Action (VLA) model: by combining visual perception, language understanding, navigation, and robotic control, VLA models provide a unified framework for developing intelligent mobile service robots capable of operating in dynamic indoor environments.

6 Roadmap and Expected Outcomes

Once the platform is integrated and the task port is validated with a working proof-of-concept application, the next major phase of development is **whole-body control**: coordinating the mobile base, telescopic column, and arms as a single kinematic and control system, rather than as separately actuated parts. This step extends the platform’s usable workspace and behavior, and is what turns it from an integrated body into a platform genuinely ready for further research.

By project completion, we expect to deliver:

- **Hardware** – mechanically complete and structurally validated.
- **Electrical** – fully wired, powered, and protected, with all sensors and actuators live.
- **Software** – fully integrated navigation, machine vision, and arm/manipulation control, running and verified.
- **Interface** – a defined, standardized set of command and feedback topics, plus a working task port, ready to receive a cognition/decision-making layer (e.g. an LLM-based planning layer) without further low-level integration work.

As this is fundamentally an R&D effort rather than a single-product build, we also intend to consolidate our design and integration methodology into a **research paper**, contributing our findings back to the research community and positioning the platform as a reusable foundation for work beyond this project.